

Abdul Basit Butt

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EDUCATION

University of Management and Technology (UMT)

B.S. Electrical Engineering

2023 – Present

Lahore, Pakistan

- **CGPA:** 3.01/4.00
- **Expected Graduation:** Spring 2027

IELTS: 7.5

RESEARCH PUBLICATIONS

AI-Based Medical Image Predictor Using CNNs for Multi-Disease Detection



Artificial Intelligence Based Control Systems for Robotics and Renewable Energy Applications



Design and Implementation of a Scalable Deep Learning-Based Cryptocurrency Price Prediction and Automated Trading



An Advanced Method for Channel Fading Parameter Estimation Based on the Generalized Gamma Distribution



Comfort Enhancement in Smartwatch Design Optimization for Continuous Elderly Healthcare Monitoring



PROJECTS

Smart Surveillance System using YOLOv8

- Developed a real-time helmet detection system using YOLOv8 trained on 5,000+ annotated images.
- Extended the model to multi-domain fire and smoke detection and analyzed dataset contamination effects.

Machine Learning-Based Intrusion Detection System

- Designed a Random Forest-based intrusion detection system for network traffic classification.
- Evaluated detection performance using accuracy, precision, recall, F1-score, and confusion matrix analysis.

Twin Rotor MIMO System (TRMS) Controller Design

- Modeled the TRMS benchmark system in MATLAB using state-space representation.
- Designed and evaluated compensators using Root Locus and classical control techniques.

Industrial Boiler Temperature Monitoring System

- Developed a MATLAB Simulink model using a Pt100 RTD sensor for industrial boiler monitoring.
- Implemented temperature visualization, alarm generation, and flame detection subsystems.

Electrical Engineering Toolkit

- Developed a web-based toolkit containing calculators for power systems, circuit analysis, transformers, and transmission lines.

TECHNICAL SKILLS

Programming : Python, MATLAB, C++, SQL

Artificial Intelligence : YOLOv8, CNNs, Random Forest, OpenCV, Computer Vision

Electrical Engineering : Control Systems, Signal Processing, Power Systems, System Modeling

Software & Tools : MATLAB, Simulink, VS Code, Git, LaTeX

Research : Technical Writing, Literature Review, Data Analysis, Scientific Publishing